Abrar Jawad Tarafder

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TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C++, SQL, PHP, PowerShell, Perl | **Frameworks:** React, Node.js, Express.js, LangChain, Streamlit, Bootstrap, NetworkX, Tkinter, HuggingFace | **Data:** PostgreSQL, MongoDB, Power BI, DAX, Tableau, Excel | **Tools & AI:** Git, GitLab CI/CD, Jira, Figma, Power Automate, REST APIs, OpenAI API, Claude Code, Ollama | **Utilities & Enterprise Systems:** GE EMS, GE Source, GE iGrid

EDUCATION

York University, Lassonde School of Engineering **Sep 2022 – Aug 2026**

Honours Bachelor of Science, Computer Science *Toronto, ON, Canada*

- **GPA:** 3.8/4.0 - Graduated with First Class Standing
- **Leadership:** Jarvis Consulting Team Lead, Microsoft Student Ambassador, Lassonde Volunteer, OUF Ambassador
- **Relevant Coursework:** Advanced OOP, Data Structures, Algorithm Design, Advanced Data Structures & Algorithms, Theory of Computer Science, Software Design, Operating Systems, Database Management Systems, Data Mining, Computer Vision, Information Networks

EXPERIENCE

York University **May 2026 – Aug 2026**

Database Developer & Research Assistant *Toronto, ON, Canada*

- Developed and maintained a **PostgreSQL research database** using **Python and SQL**, building automated **ETL pipelines** to clean, integrate, and standardize **10,000+ historical records**, improving data integrity and accessibility for academic research.
- Automated data extraction from **28 archival PDFs** using **Python**, transforming unstructured personnel records into normalized datasets and integrating **18+ role definitions** to expand research data coverage.

Hydro One **May 2024 - Aug 2025**

OT Transmissions Application Intern, Co-op *Barrie, ON, Canada*

- Supported **GE EMS** applications and biweekly/monthly **QA database releases**, constantly maintaining, automating and updating development environments/equipment data and collaborating on an in-house release management application using **ReactJS, Bootstrap, and PostgreSQL**.
- Automated **GitLab CI/CD workflows** using **Perl and PowerShell**, streamlining cross-domain server transfers and reducing shuttle processing time by **73%**.
- Automated the **GE EMS LiveSnap loading process** using GitLab pipelines and Perl scripts, reducing monthly QA database loading time by **87%**.
- Developed **Python and PowerShell** scripts to automate GitLab error parsing and XML conversion, reducing external display pipeline processing time from **8 hours to 5 minutes**.
- Developed **SQL and PL/SQL validation scripts** within **GE Source** to automate field checks, trigger error alerts, and improve data accuracy for the Application Modelling team.
- Supported an **asset acquisition project** by mapping equipment records, tracking nameplate documentation and in-service dates, and maintaining accurate asset data.

WalkWithWeb Inc. **May 2023 – May 2024**

Web Developer & Research Assistant *Toronto, ON, Canada*

- Collaborated with **3 departments** to gather requirements and develop **responsive websites** using HTML, CSS, Bootstrap, and JavaScript, aligning solutions with stakeholder needs.
- Designed **interactive Figma prototypes** and analyzed **social media engagement** to guide UI/UX and marketing decisions, contributing to a **50% increase in content output**.

PROJECTS

Cruise Critic AI Analytics and Power BI Project | *Python, SQL, Power BI, DAX, Ollama, Streamlit, DuckDB*

- Designed a **normalized SQL database and star schema** with **3 Power BI dashboards** analyzing cruise pricing, ratings, and trends. Built an **AI analytics assistant** using **Ollama, Python, Streamlit, and DuckDB** for natural-language queries, validated SQL analysis, and interactive visualizations.

Atlas AI Research Workspace Project | *Python, LangChain, OpenAI API, Streamlit*

- Developed an **AI research agent** with **multi-step tool calling, web search, calculator integration, and conversational memory**, generating structured research briefs with verified source URLs, interactive follow-ups, and PDF export.

Freshroute Grocery Platform Project | *React, Node.js, Express.js, MongoDB*

- Collaborated in an **Agile development team** to build a **full-stack grocery platform** with REST APIs, MongoDB integration, product browsing, shopping cart, checkout, and recipe-based ingredient scaling.

TTC Network Analysis Project | *Python, NetworkX, Dijkstra, A*, PageRank*

- Modeled **9,406 TTC nodes** as a **weighted directed graph** using GTFS data, applying shortest-path algorithms, PageRank, centrality metrics, and disruption simulations to evaluate routing efficiency and network resilience.